

SYNOPSIS CHAMPIONSHIP PROJECT ABSTRACT

The Abstract is a required part of your project. Bring 1 copy at Project Check-in.

Your abstract should be written after you finish the research and experimentation and should include:

- **Project number**, title and the full name of every team member
- Purpose of the project
- Procedure
- Observations/data/results
- Conclusions
- 250 words or less (to fit on one page)
- Use a black font size 10 to 12

P10: Personalized Deep Learning Model for Closed-Loop 40 Hz Entrainment to Optimize Theta-Gamma Coupling in Alzheimer's

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Sound pulses at 40 Hz synchronize brain gamma oscillations and activate immune cells that clear amyloid plaques, common in Alzheimer's disease. Clinical protocols use rigid fixed schedules that ignore individual brain responses. Half of patients habituate within minutes; the other half maintain strong neural coupling. A one-size-fits-all approach serves neither group. This project asks: can we predict when a patient's brain will lose entrainment, and intervene before it happens? I analyzed EEG from 35 dementia patients (OpenNeuro ds005048), computing phase-amplitude coupling (PAC) between theta and gamma rhythms as an entrainment biomarker. I trained a causal Temporal Convolutional Network (31,000 parameters, dilations [1,2,4,8]) to forecast PAC 5-10 seconds ahead. At 1–2 seconds, simple baselines suffice; at 5–10 seconds—where a controller must act—all baselines collapse to negative R^2 while the TCN maintains R^2 of 0.25, a +0.5 margin. Validated on all 35 subjects' real EEG, the predictive controller achieved 72.1% alignment versus 64.5% reactive (Hedges' $g = 1.31$, $p < 0.001$). It directed stimulation to 82.6% of low-PAC windows versus 51.7% ($g = 4.47$, $p < 0.001$), reaching 91% of the theoretical oracle. Every patient benefited (binomial $p < 0.001$). Adaptive efficiency gains of +9-11% held across six fatigue levels and four different habituation model assumptions (all $p < 10^{-13}$), demonstrating robust personalized control.